

COMP3013 DBMS Lab 7

United International College

Data Modification

- In the last lab, we have created new tables for UIC database.
- Check “UIC definition.txt” for the sample code and compare it with your query. If you have any question, please ask your instructor or TA.
- In this lab, we will insert, delete, and update data.

Note:

- Usually, database users are only allowed to change data (table rows), which is covered by this lab.
- Table columns are related to the logical design and managed by database managers.
- To change the logical design, you need **ALTER TABLE** (last lab).

Insert

- To insert a record for a program:
 - Program code: 1001
 - Program name: Computer Science
 - Division: Science and Technology
 - Director: NULL (unknown)

```
INSERT INTO program (p_code, p_name, division, director_id)
VALUES (1001, 'Computer Science', 'Science and Technology', NULL);
```

- The **schema** of the table is to remind users the data types, which can be omitted.
- Currently, the instructor table is empty. So, the program director is **unknown**.

Example

- Insert the information of a new instructor,
 - ID: 20001
NAME: S.T. Kwok
Title: Professor
Salary: 100000
Program: Computer Science

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 - ID: 20001
NAME: S.T. Kwok
Title: Professor
Salary: 100000
Program: Computer Science

```
INSERT INTO instructor VALUES  
(20001, 'S.T. Kwok', 'Professor', 1001);
```

Example

- Try to insert
 - Course:
Name: Introduction to Biology, Credits: 3,
Domain: ENVS, Course Number: 2001
 - Instructor:
ID: 20002, Name: S.H. Zee, Title: Professor,
Salary: 100000, Program: Environmental Science
 - Program:
ID: 1002, Name: Environmental Science,
Division: Science and Technology, Program director: Unknown
- Why the instructor cannot be inserted before the program?

Example

```
INSERT INTO course VALUES  
('Introduction to Biology', 3, 'ENVS', 2001);
```

```
INSERT INTO program VALUES  
(1002, 'Environmental Science', 'Science and Technology', NULL);
```

```
INSERT INTO program VALUES  
(20002, 'S.H. Zee', 'Professor', 100000, 1002);
```

- Why the instructor cannot be inserted before the program?
You cannot sign a professor to a program which does not exist.

Insert

- Sometimes we can insert data constructed from other tables.
- Suppose we want to express that the program Environmental Science offers the course Introduction to Biology.

```
INSERT INTO offer
```

```
SELECT p_code,c_name  
FROM program,course  
WHERE p_name='Environmental Science'  
      AND c_name = 'Introduction to Biology'
```

- **SELECT...FROM...WHERE...** constructs the tuples which are inserted to “offer”.

Update

- Suppose Prof. S.T. Kwok becomes the new program director for Computer Science. We need to update the table.

```
UPDATE program
SET director_id = (SELECT id FROM instructor WHERE i_name='S.T. Kwok')
WHERE p_name = 'Computer Science'
```

- **UPDATE**: the keyword to update a table, followed by the table name.
- **SET**: the keyword to update a specific column, followed by an assignment operation.
- **(...)**: a **subquery** to find the person id of S.T. Kwok.
- **WHERE**: only update the rows which satisfy the condition in the WHERE clause.
- A subquery is a query inside another query. More about this will be in the next lab.

Delete

- Suppose Prof. S.T. Kwok has retired. All his information should be removed from the system.

```
DELETE FROM instructor WHERE i_name='S.T. Kwok'
```

- **Every row** which satisfies the **condition** will be removed. Thus, **be careful**.
- The system shows an error when the query is executed. The reason is same as the example on page 5.
- Thus, change the program director for Computer Science to NULL first. Then, remove Prof. S.T. Kwok.

Exercises

Implement the logical design in MySQL

- Insert all data in the package
- To insert multiple rows into one table at a time, try

```
INSERT INTO table VALUES (t1),(t2),...
```

- The student whose GPA is below 1.7 will be disqualified from the college. Find such students and remove their information from the database.

Save your queries in a txt file. Rename it as “COMP3013 Lab7 ###.txt”, where “###” is your student ID. And submit it on iSpace. The DDL is 24 hours after the lab.